SOC 205B Data Analysis in Sociology. Problem Set 1. Interactions. Arenas, Spring 2024
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1. $prestg10 = \beta_0 + \beta_1 yreducmale + \beta_2 age + \beta_3 yreducmale * age + \varepsilon$

2. $\frac{\Delta \widehat{prestg}10_i}{\Delta age_i} = \frac{\Delta(40.43 + 0.13 yreducmale_i + 0.07 age_i + 0.0003 yreducmale_i * age_i)}{\Delta age_i}$

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. reg prestg10 yreducmale age yreducmale*age

      Source |       SS           df       MS      Number of obs   =     3,135
-------------+----------------------------------   F(3, 3131)      =     16.75
       Model |  9175.59974         3  3058.53325   Prob > F        =    0.0000
    Residual |  571877.792     3,131  182.650205   R-squared       =    0.0158
-------------+----------------------------------   Adj R-squared   =    0.0148
       Total |  581053.392     3,134  185.403124   Root MSE        =    13.515

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     prestg10 | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
   yreducmale |   .1302136   .1003798     1.30   0.195    -.0666034    .3270305
          age |   .0717013   .0181433     3.95   0.000     .0361273    .1072753
yreducmaleage |   .0003391   .0018816     0.18   0.857    -.0033503    .0040285
        _cons |   40.43383   .9595981    42.14   0.000     38.55232    42.31533
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3. $\frac{\Delta \widehat{prestg}10_i}{\Delta age_i * \Delta male_i} = \frac{\Delta(10.50 + 2.41 yreduc_i + 0.07 agemale_i + (-0.004) yreduc_i * agemale_i)}{\Delta age_i * \Delta male_i}$

$$\frac{\Delta \widehat{prestg}10_i}{\Delta age_i * \Delta male_i} = 0.07 + (-0.004) yreduc$$

The partial effect of being of male gender and aged (agemale) over the occupational prestige score (prestg10) is incremented by 0.07 for each year of education attended (yreduc) in a population of Mexican migrants aged over 18 years residing in 2022 in the U.S.

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. reg prestg10 yreduc agemale yreducagemale

      Source |       SS           df       MS      Number of obs   =     3,135
-------------+----------------------------------   F(3, 3131)      =    320.55
       Model |  136531.682         3  45510.5607   Prob > F        =    0.0000
    Residual |   444521.71     3,131  141.974356   R-squared       =    0.2350
-------------+----------------------------------   Adj R-squared   =    0.2342
       Total |  581053.392     3,134  185.403124   Root MSE        =    11.915

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     prestg10 | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       yreduc |   2.412581   .0983833    24.52   0.000     2.219679    2.605483
      agemale |   .0756149   .0385101     1.96   0.050     .0001072    .1511225
yreducagemale |    -.00441   .0026501    -1.66   0.096    -.0096062    .0007862
        _cons |   10.50391   1.424563     7.37   0.000     7.710742    13.29708
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4. $\frac{\Delta \widehat{prestg}10_i}{\Delta yreduc_i * \Delta male_i} = \frac{\Delta(40.43 + 0.72age_i + 0.13yreducmale_i + 0.003yreducmale_i * age_i)}{\Delta yreduc_i * \Delta male_i}$

$$\frac{\Delta \widehat{prestg}10_i}{\Delta yreduc_i * \Delta male_i} = 0.13 + 0.0003age_i$$

The partial effect of being of male gender and have also higher number of years of education attendance *(yreducmale)* over the occupational prestige score *(prestg10*) is incremented by 0.13 for each year old (*age*) accumulated counting from 19 years aged in a population of Mexican migrants aged over 18 years residing in 2022 in the U.S.

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. reg prestg10 age yreducmale yreducagemale

      Source |       SS           df       MS      Number of obs   =     3,135
-------------+----------------------------------   F(3, 3131)      =     16.75
       Model |  9175.59974         3  3058.53325   Prob > F        =    0.0000
    Residual |  571877.792     3,131  182.650205   R-squared       =    0.0158
-------------+----------------------------------   Adj R-squared   =    0.0148
       Total |  581053.392     3,134  185.403124   Root MSE        =    13.515

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     prestg10 | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
          age |   .0717013   .0181433     3.95   0.000     .0361273    .1072753
   yreducmale |   .1302136   .1003798     1.30   0.195    -.0666034    .3270305
yreducagemale |   .0003391   .0018816     0.18   0.857    -.0033503    .0040285
        _cons |   40.43383   .9595981    42.14   0.000     38.55232    42.31533
------------------------------------------------------------------------------
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5. $\widehat{prestg}10_i = 40.43 + 0.72age_i + 0.13yreducmale_i + 0.003yreducmale_i * age_i$

6.1. $prestg10 = \beta_0 + \beta_1 yreduc + \beta_2 male + \beta_3 yreduc * male + \varepsilon$

6.2.

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. gen yreducmale = yreduc*male
(24 missing values generated)

. *Define the model for the regression considering the interaction.
. reg prestg10 yreduc male yreduc*male

      Source |       SS           df       MS      Number of obs   =     3,302
-------------+----------------------------------   F(3, 3298)      =    334.50
       Model |  141938.345         3  47312.7818   Prob > F        =    0.0000
    Residual |  466483.921     3,298  141.444488   R-squared       =    0.2333
-------------+----------------------------------   Adj R-squared   =    0.2326
       Total |  608422.267     3,301  184.314531   Root MSE        =    11.893

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    prestg10 | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      yreduc |   2.370157   .0994127    23.84   0.000     2.175241    2.565074
        male |   2.484912   2.098483     1.18   0.236    -1.629549    6.599373
  yreducmale |  -.1636225    .145187    -1.13   0.260    -.4482884    .1210433
       _cons |   11.28315   1.444982     7.81   0.000     8.449995     14.1163
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. *For confirmation of results of strategy 2 in question 6 in comparison with strategy 1.
. reg prestg10 yreduc if male==1

      Source |       SS           df       MS      Number of obs   =     1,549
-------------+----------------------------------   F(1, 1547)      =    428.47
       Model |  61507.7435         1  61507.7435   Prob > F        =    0.0000
    Residual |  222074.653     1,547  143.551812   R-squared       =    0.2169
-------------+----------------------------------   Adj R-squared   =    0.2164
       Total |  283582.396     1,548  183.192763   Root MSE        =    11.981

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    prestg10 | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      yreduc |   2.206535   .1065983    20.70   0.000     1.997442    2.415627
       _cons |   13.76806   1.533022     8.98   0.000     10.76104    16.77508
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. reg prestg10 yreduc if male==0

      Source |       SS           df       MS      Number of obs   =     1,753
-------------+----------------------------------   F(1, 1751)      =    576.00
       Model |  80400.0777         1  80400.0777   Prob > F        =    0.0000
    Residual |  244409.269     1,751  139.582678   R-squared       =    0.2475
-------------+----------------------------------   Adj R-squared   =    0.2471
       Total |  324809.346     1,752  185.393462   Root MSE        =    11.815

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    prestg10 | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      yreduc |   2.370157   .0987563    24.00   0.000     2.176465     2.56385
       _cons |   11.28315   1.435441     7.86   0.000      8.46779    14.09851
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6.3.

	Model 1			Model 2		
	Coefficient	*Std. err.*	*P>t*	*Coefficient*	*Std. err.*	*P>t*
yreduc	2.293444	0.0724554	0	2.370157	0.0988202	0
male	0.1666502	0.4149017	0.688	0.1167348	1.587466	0.941
yreducmale						
2				11.8598	12.04346	0.325
3				7.489645	5.735198	0.192
4				21.61949	6.289547	0.001
5				0.7493307	12.00326	0.95
6				9.74281	4.091137	0.017
7				6.009016	5.632108	0.286
8				8.138858	3.501293	0.02
9				5.227034	3.020383	0.084
10				0.1147599	2.621365	0.965
11				-0.4928903	2.425348	0.839
12				-0.6363256	1.75743	0.717
13				-1.049535	1.946969	0.59
14				-0.8891035	1.755613	0.613
15				-4.21891	1.978557	0.033
16				0.7462267	1.652457	0.652
17				2.898351	2.079216	0.163
18				1.607873	1.949605	0.41
19				-0.2384285	2.45147	0.923
20				0	(omitted)	
_cons	12.37643	1.070964	0	11.28315	1.436369	0

6.4. *if*: $prestg10_i = 11.28 + 2.37yeduc + 2.48male + (-0.16)yreducmale$

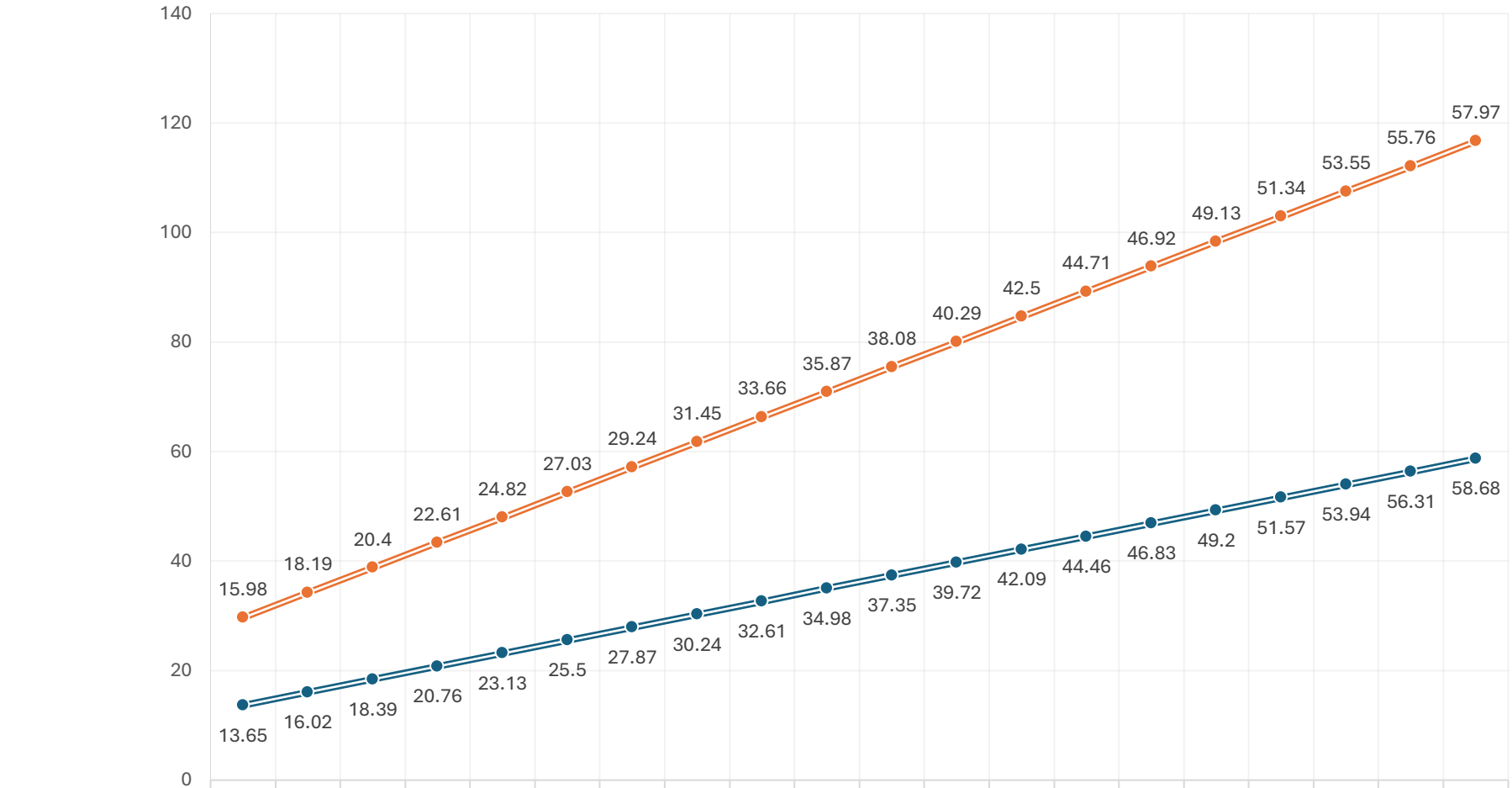
$$\widehat{prestg10}_f = 11.28 + 2.37yeduc$$

$$\widehat{prestg10}_m = 13.77 + 2.21yeduc$$

6.4. For women (f, male==0), an increase of one year of education attendance (yreduc) increases in 2.37 points its score in the scale of occupational prestige (prestg10).

6.6. For men (m, male==1), an increase of one year of education attendance (yreduc) increases in 2.21 points its score in the scale of occupational prestige (prestg10).

6.7.



	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
prestg10_male	15.98	18.19	20.4	22.61	24.82	27.03	29.24	31.45	33.66	35.87	38.08	40.29	42.5	44.71	46.92	49.13	51.34	53.55	55.76	57.97
prestg10_female	13.65	16.02	18.39	20.76	23.13	25.5	27.87	30.24	32.61	34.98	37.35	39.72	42.09	44.46	46.83	49.2	51.57	53.94	56.31	58.68

7.1. $prestg10 = \beta_0 + \beta_1 yreduc + \beta_2 age + \beta_3 yreduc * age + \varepsilon$

7.2.

```
. reg prestg10 yreduc age yreducage

      Source |       SS           df       MS      Number of obs   =     3,140
-------------+----------------------------------   F(3, 3136)      =    330.36
       Model |  139673.136         3  46557.7119   Prob > F        =    0.0000
    Residual |  441958.486     3,136   140.93064   R-squared       =    0.2401
-------------+----------------------------------   Adj R-squared   =    0.2394
       Total |  581631.621     3,139  185.292011   Root MSE        =    11.871

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    prestg10 | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      yreduc |   2.312176   .2361681     9.79   0.000     1.849117    2.775236
         age |   .0672964   .0619433     1.09   0.277     -.054157    .1887499
   yreducage |  -.0003749   .0043552    -0.09   0.931    -.0089143    .0081644
       _cons |   9.142413   3.349016     2.73   0.006     2.575927     15.7089
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7.3.

	Model 1 (restricted)			Model 2 (nested)		
	Coefficient	*Std. err.*	*P>t*	*Coefficient*	*Std. err.*	*P>t*
yreduc	2.268	0.076	0.000	-0.576	3.546	0.871
age				-0.254	0.750	0.735
_Iage_19	2.533	4.178	0.544	-5.076	53.585	0.925
_Iage_20	3.102	3.793	0.414	-10.966	43.449	0.801
_Iage_21	4.256	3.760	0.258	-17.421	41.048	0.671
_Iage_22	0.188	3.880	0.961	-13.831	40.689	0.734
_Iage_23	3.341	3.740	0.372	-24.433	39.296	0.534
_Iage_24	6.319	3.761	0.093	-18.820	39.343	0.632
_Iage_25	4.499	3.760	0.232	-15.332	37.600	0.683
_Iage_26	6.451	3.684	0.080	-29.917	37.331	0.423
_Iage_27	5.788	3.764	0.124	-32.502	37.238	0.383
_Iage_28	6.971	3.664	0.057	-23.790	36.064	0.510
_Iage_29	6.034	3.639	0.097	-16.297	35.418	0.645
_Iage_30	3.739	3.649	0.306	-25.401	35.706	0.477
_Iage_31	8.235	3.618	0.023	-27.625	34.578	0.424
_Iage_32	8.912	3.575	0.013	-30.687	33.796	0.364
_Iage_33	8.487	3.606	0.019	-14.777	33.384	0.658
_Iage_34	4.960	3.649	0.174	-28.053	32.987	0.395
_Iage_35	8.874	3.649	0.015	-16.409	32.590	0.615
_Iage_36	7.684	3.614	0.034	-28.632	32.251	0.375
_Iage_37	6.688	3.719	0.072	-31.072	31.887	0.330
_Iage_38	7.186	3.658	0.050	-25.474	31.651	0.421
_Iage_39	9.100	3.592	0.011	-28.139	31.264	0.368
_Iage_40	9.193	3.635	0.011	-13.689	30.809	0.657
_Iage_41	6.137	3.726	0.100	-5.805	30.612	0.850
_Iage_42	9.146	3.607	0.011	-33.942	30.570	0.267
_Iage_43	4.791	3.766	0.203	-25.776	30.706	0.401
_Iage_44	10.173	3.675	0.006	-8.126	29.257	0.781
_Iage_45	7.981	3.672	0.030	-4.933	29.291	0.866
_Iage_46	8.246	3.693	0.026	-17.709	29.173	0.544
_Iage_47	12.862	3.688	0.000	10.164	28.378	0.720
_Iage_48	5.169	3.742	0.167	-25.919	28.674	0.366
_Iage_49	10.104	3.683	0.006	-14.791	27.987	0.597
_Iage_50	7.978	3.795	0.036	-13.166	27.783	0.636
_Iage_51	8.034	3.682	0.029	-16.168	27.399	0.555
_Iage_52	8.116	3.748	0.030	-17.597	28.319	0.534
_Iage_53	9.598	3.810	0.012	1.803	27.942	0.949
_Iage_54	8.881	3.716	0.017	-11.018	27.163	0.685
_Iage_55	7.266	3.632	0.046	-22.579	27.040	0.404
_Iage_56	5.053	3.671	0.169	-0.314	27.101	0.991
_Iage_57	7.244	3.690	0.050	-20.109	27.028	0.457
_Iage_58	8.876	3.763	0.018	-16.652	27.290	0.542
_Iage_59	7.960	3.690	0.031	-13.659	27.001	0.613
_Iage_60	7.885	3.576	0.028	-16.342	26.896	0.544

_Iage_61	7.457	3.666	0.042	-18.697	26.913	0.487
_Iage_62	8.440	3.637	0.020	-40.512	27.418	0.140
_Iage_63	8.501	3.726	0.023	-13.091	28.149	0.642
_Iage_64	9.198	3.697	0.013	-16.014	27.045	0.554
_Iage_65	6.413	3.613	0.076	-2.673	27.565	0.923
_Iage_66	7.083	3.597	0.049	-24.662	27.435	0.369
_Iage_67	9.194	3.763	0.015	-15.595	28.017	0.578
_Iage_68	6.226	3.592	0.083	-9.810	27.447	0.721
_Iage_69	7.306	3.663	0.046	-11.651	27.947	0.677
_Iage_70	6.981	3.616	0.054	-8.036	27.810	0.773
_Iage_71	9.234	3.786	0.015	13.071	29.473	0.657
_Iage_72	8.315	3.666	0.023	-3.077	28.970	0.915
_Iage_73	9.008	3.887	0.021	-12.651	29.581	0.669
_Iage_74	11.290	3.756	0.003	-33.001	29.484	0.263
_Iage_75	7.373	3.882	0.058	-4.125	30.155	0.891
_Iage_76	10.548	3.981	0.008	-8.083	30.305	0.790
_Iage_77	8.200	3.916	0.036	-16.560	31.089	0.594
_Iage_78	12.861	4.052	0.002	-25.235	32.677	0.440
_Iage_79	9.524	3.940	0.016	-22.681	31.923	0.477
_Iage_80	11.572	4.142	0.005	0.346	31.628	0.991
_Iage_81	9.611	4.650	0.039	-34.699	34.057	0.308
_Iage_82	9.640	4.563	0.035	-6.176	34.816	0.859
_Iage_83	1.202	4.647	0.796	-0.211	34.873	0.995
_Iage_84	11.442	4.222	0.007	-7.901	34.119	0.817
_Iage_85	14.916	5.325	0.005	6.853	38.492	0.859
_Iage_86	13.627	4.979	0.006	6.088	34.882	0.861
_Iage_87	7.488	6.229	0.229	26.506	37.842	0.484
_Iage_88	12.858	6.772	0.058	0.000	(omitted)	
_Iage_89	8.185	4.311	0.058	3.853	36.768	0.917
yreduc				0.000	(omitted)	
_IageXyred_19				0.856	4.575	0.852
_IageXyred_20				1.496	3.807	0.694
_IageXyred_21				2.110	3.679	0.566
_IageXyred_22				1.569	3.687	0.670
_IageXyred_23				2.598	3.638	0.475
_IageXyred_24				2.426	3.668	0.508
_IageXyred_25				2.032	3.602	0.573
_IageXyred_26				3.264	3.607	0.365
_IageXyred_27				3.413	3.626	0.347
_IageXyred_28				2.908	3.593	0.418
_IageXyred_29				2.337	3.586	0.515
_IageXyred_30				2.830	3.628	0.435
_IageXyred_31				3.317	3.591	0.356
_IageXyred_32				3.630	3.580	0.311
_IageXyred_33				2.466	3.580	0.491
_IageXyred_34				3.185	3.583	0.374
_IageXyred_35				2.644	3.585	0.461
_IageXyred_36				3.429	3.585	0.339

_IageXyred_37	3.530	3.584	0.325
_IageXyred_38	3.212	3.593	0.371
_IageXyred_39	3.535	3.591	0.325
_IageXyred_40	2.576	3.589	0.473
_IageXyred_41	1.850	3.594	0.607
_IageXyred_42	3.962	3.603	0.272
_IageXyred_43	3.148	3.620	0.385
_IageXyred_44	2.304	3.583	0.520
_IageXyred_45	1.981	3.592	0.581
_IageXyred_46	2.891	3.597	0.422
_IageXyred_47	1.184	3.583	0.741
_IageXyred_48	3.299	3.602	0.360
_IageXyred_49	2.872	3.583	0.423
_IageXyred_50	2.617	3.584	0.465
_IageXyred_51	2.859	3.576	0.424
_IageXyred_52	2.973	3.606	0.410
_IageXyred_53	1.746	3.605	0.628
_IageXyred_54	2.610	3.581	0.466
_IageXyred_55	3.333	3.582	0.352
_IageXyred_56	1.651	3.585	0.645
_IageXyred_57	3.196	3.588	0.373
_IageXyred_58	3.078	3.598	0.392
_IageXyred_59	2.819	3.590	0.432
_IageXyred_60	3.026	3.586	0.399
_IageXyred_61	3.160	3.581	0.378
_IageXyred_62	4.864	3.606	0.177
_IageXyred_63	2.888	3.622	0.425
_IageXyred_64	3.176	3.585	0.376
_IageXyred_65	2.055	3.597	0.568
_IageXyred_66	3.670	3.590	0.307
_IageXyred_67	3.184	3.604	0.377
_IageXyred_68	2.581	3.580	0.471
_IageXyred_69	2.812	3.589	0.433
_IageXyred_70	2.546	3.578	0.477
_IageXyred_71	1.264	3.631	0.728
_IageXyred_72	2.353	3.600	0.513
_IageXyred_73	3.059	3.609	0.397
_IageXyred_74	4.601	3.599	0.201
_IageXyred_75	2.396	3.619	0.508
_IageXyred_76	2.911	3.607	0.420
_IageXyred_77	3.368	3.635	0.354
_IageXyred_78	4.337	3.690	0.240
_IageXyred_79	3.866	3.631	0.287
_IageXyred_80	2.418	3.628	0.505
_IageXyred_81	4.726	3.688	0.200
_IageXyred_82	2.822	3.732	0.450
_IageXyred_83	1.818	3.707	0.624
_IageXyred_84	3.111	3.657	0.395

_IageXyred_85				2.350	3.818	0.538
_IageXyred_86				2.214	3.685	0.548
_IageXyred_87				-0.126	3.874	0.974
_IageXyred_88				2.725	4.211	0.518
_IageXyred_89				2.091	3.696	0.572
_cons	5.258	3.394	0.121	42.213	51.634	0.414

7.4.

$$\widehat{prestg}10_i = 9.14 + 2.31yreduc_i + (-0.003)yreduc_i * age_i$$

7.5.

$$\frac{\Delta\widehat{prestg}10_i}{\Delta age_i} = \frac{\Delta(9.14 + 2.31yreduc_i + 0.07age_i + (-0.0003)yreduc_i * age_i)}{\Delta age_i}$$

$$\frac{\Delta\widehat{prestg}10_i}{\Delta age_i} = 0.67 + (-0.0003)yreduc_i$$

The partial effect of being aged *(age)* over the occupational prestige score (*prestg10*) is incremented by 0.67 for each year of education *attended (yreduc*) in a population of Mexican migrants aged over 18 years residing in 2022 in the U.S.

Based on the table presented above (Model 2 in 7.3), none of each age categories over 18 years old are significant in its partial effect over the occupational prestige score (*prestg10*).

7.6.

The Wald test reject the null hypothesis of no improvement between the coefficients of *yreduc* and *yreducage* as the comparison between the restricted model and the extended model. The result states that the difference between the two models is significant at a significance level of $p<0.0001$, stating the rejection of the null hypothesis because the coefficients are jointly significant different from 0.

The difference between the partial effect of yreduc over prestg10 is significant when the interaction with the control variable age yreducage is introduced. So, the partial effect of being aged over the occupational prestige score (*prestg10*) of an increment in 0.67 by each year of education *(yreduc*) is significant in a population of Mexican migrants aged over 18 years residing in 2022 in the U.S.

```
. *question 7. generate yreducage
. gen yreducage = yreduc*age
(186 missing values generated)

. reg prestg10 yreduc age yreducage

      Source |       SS           df       MS      Number of obs   =     3,140
-------------+----------------------------------   F(3, 3136)      =    330.36
       Model |  139673.136         3  46557.7119   Prob > F        =    0.0000
    Residual |  441958.486     3,136   140.93064   R-squared       =    0.2401
-------------+----------------------------------   Adj R-squared   =    0.2394
       Total |  581631.621     3,139  185.292011   Root MSE        =    11.871

------------------------------------------------------------------------------
    prestg10 | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      yreduc |   2.312176   .2361681     9.79   0.000     1.849117    2.775236
         age |   .0672964   .0619433     1.09   0.277     -.054157    .1887499
   yreducage |  -.0003749   .0043552    -0.09   0.931    -.0089143    .0081644
       _cons |   9.142413   3.349016     2.73   0.006     2.575927     15.7089
------------------------------------------------------------------------------

. testparm yreduc yreducage

 ( 1)  yreduc = 0
 ( 2)  yreducage = 0

       F(  2,  3136) =  476.17
            Prob > F =    0.0000
```